

$$= \int \frac{d^d \omega}{\omega_0} K_{\mu i} \omega_0^2 K_{\Delta}(\omega, x) \overset{2}{\partial}_{\mu} K_{\Delta}$$

$$\frac{\partial}{\partial \omega_{\mu}} K_{\Delta}(\omega, x) = \frac{\partial}{\partial \omega_{\mu}} \left(\frac{\omega_0}{\omega_0^2 + (\bar{\omega} - \bar{x})^2} \right)^{\Delta}$$

$$\frac{\partial}{\partial \omega_0} \left(\frac{\omega_0^{\Delta-1}}{\omega_0^2 + (\bar{\omega} - \bar{x})^2} \right)^{\Delta} = \Delta \frac{\omega_0^{\Delta-1}}{\omega_0^2 + (\bar{\omega} - \bar{x})^2} - \Delta \frac{\omega_0^{\Delta}}{(\omega_0^2 + (\bar{\omega} - \bar{x})^2)^{\Delta+1}} \cdot 2\omega_0$$

$$= \Delta K_{\Delta}(\omega, x) \left(\frac{1}{\omega_0} - \frac{2\omega_0}{(\omega_0^2 + (\bar{\omega} - \bar{x})^2)} \right)$$

$$= \Delta K_{\Delta}(\omega, x) \left(\frac{\omega_0^2 + (\bar{\omega} - \bar{x})^2 - 2\omega_0^2}{\omega_0(\omega_0^2 + (\bar{\omega} - \bar{x})^2)} \right)$$

$$= \Delta K_{\Delta} \left(\frac{-\omega_0^2 + (\bar{\omega} - \bar{x})^2}{\omega_0(\omega_0^2 + (\bar{\omega} - \bar{x})^2)} \right)$$

$$\frac{\partial}{\partial \omega_i} \left(\frac{\omega_0^{\Delta} (\bar{\omega} - \bar{x})_i (-\Delta)}{(\omega_0^2 + (\bar{\omega} - \bar{x})^2)^{\Delta+2}} \right) = K_{\Delta} \frac{\Delta (\bar{\omega} - \bar{x})_i}{(\omega_0^2 + (\bar{\omega} - \bar{x})^2)^2}$$

$$\Rightarrow \int \frac{d^d \omega}{\omega_0^{\Delta+1}} \left(\frac{\omega_0^{\Delta-2}}{(\omega_0^2 + (\bar{\omega} - \bar{x})^2)^{\Delta-1}} \right) \delta_{\mu i} (\omega - z) \left(\frac{\partial K_{\Delta}(\omega, \bar{x})}{\partial \omega_{\mu}} K_{\Delta}(\omega, \bar{y}) - K_{\Delta} \frac{\partial K_{\Delta}}{\partial \omega_{\mu}} \right)$$

• Set $\bar{z} \rightarrow 0$ i.e. $x \rightarrow x-z$ and $y \rightarrow y-z$

$$K_{\mu i}(\omega', 0) = \frac{1}{\omega_0'} \left(\frac{\omega_0'}{\omega'^2} \right)^{\Delta-1} = \frac{\omega'^2 \omega^{2(\Delta-1)}}{\omega_0} \left(\frac{\omega_0}{\omega'^2} \right)^{\Delta-1}$$

$$K_{\mu i}(\omega', 0) = K_{\mu i}(\omega_0) \omega'^{2\Delta}$$

$$\text{Since } I_{\mu i}(\omega') = I_{\mu i}(\omega)$$

• Derivatives:

$$\frac{\partial}{\partial \omega_{\mu}} = \frac{\partial \omega'_0}{\partial \omega_{\mu}} \frac{\partial}{\partial \omega'_0} = \frac{1}{(\omega')^2} I_{\sigma}^{\mu}(\omega') \frac{\partial}{\partial \omega'_0}$$

$$= -i \int \frac{d^{d+1} \omega'}{(\omega'_0)^{d+1}} \frac{(\omega'_0)^{\Delta-1}}{\omega_0} I_{\mu i} I_{\sigma}^{\mu} (K \dots)$$

$$= -i \int \frac{d^{d+1} \omega'}{(\omega'_0)^2} \left(\frac{\omega_0}{\omega'^2} \right) \frac{1}{|x|^{2\Delta}} \frac{1}{|y|^{2\Delta}} \delta_{\sigma i} K_{\Delta}(\omega', \bar{x}') \frac{\partial}{\partial \omega'_0} K_{\Delta}(\omega', \bar{y}')$$

$$\delta_{\sigma i} \rightarrow \delta_{ii}$$

$$\frac{\partial}{\partial x'_i} (K_{\Delta}(\omega', x')) = \Delta \frac{(\bar{\omega}' - \bar{x}')_i}{\omega_0'} K_{\Delta}$$

$$(\omega_0'^2 + (\bar{\omega}' - \bar{x}')^2)^{-1}$$

$$= -i \int \frac{d^{d+1}\omega'}{(\omega_0')^2} \left(\frac{\omega_0}{\omega'} \right)^{\Delta} \frac{1}{|x|^{2\Delta}} \frac{1}{|y|^{2\Delta}} (A) \left[\frac{(\bar{\omega}' - \bar{x}')^2}{(\omega_0'^2 + (\bar{\omega}' - \bar{x}')^2)^2} K_{\Delta}(\omega', x') K_{\Delta}(\omega', y') \right.$$

$$\left. - \frac{(\bar{\omega}' - \bar{y}')^2}{(\omega_0'^2 + (\bar{\omega}' - \bar{y}')^2)^2} K_{\Delta}(\omega', x') K_{\Delta}(\omega', y') \right]$$

$$= -i \frac{1}{|x|^{2\Delta} |y|^{2\Delta}} \int \frac{d^{d+1}\omega'}{\omega'^2} \frac{(\omega_0')^{2\Delta-1}}{(\bar{\omega}' - \bar{x}')^{2\Delta} (\bar{\omega}' - \bar{y}')^{2\Delta}} \left[\frac{\bar{\omega}' - \bar{x}'}{(\bar{\omega}' - \bar{x}')^{\Delta}} - \frac{\bar{\omega}' - \bar{y}'}{(\bar{\omega}' - \bar{y}')^{\Delta}} \right]$$

(a) $\int \frac{d^{d+1}\omega}{\omega'^2} (\omega_0')^{2\Delta-1} \int dx_1 dx_2 \delta(x_1 + x_2 - 1) \left((\bar{\omega}' - \bar{x}')^{2(\Delta+2)} - (\bar{\omega}' - \bar{y}')^{2(\Delta+2)} \right)$

$$= -i \int \frac{d^d \omega' d\omega_0'}{\omega'^2} \int_0^1 d\alpha \left(\alpha (\bar{\omega}' - \bar{x}')^{2(\Delta+2)} + (1-\alpha) (\bar{\omega}' - \bar{y}')^{2(\Delta+2)} \right)^{-2} \omega_0'^{2\Delta-1}$$

$$= -i \int \frac{d^{d+1}\omega}{(\omega_0')^2} (\omega_0')^{2\Delta-1} \int_0^1 d\alpha \left(\alpha (\bar{\omega}' - \bar{x}')^{2(\Delta+2)} - (\bar{\omega}' - \bar{y}')^{2\Delta} \right)^{-2}$$

$$\bar{q}_i \equiv (\omega' - \bar{x}'); \quad = |\bar{\omega}' - \bar{x}'|;$$

only the $\bar{x}'$ and $\bar{y}'$ terms remain since

$$\int d^d \omega: f(\omega^2) = 0$$

Go Back to the 3pt fu =

$$\int \frac{d\omega_0 d^d \bar{\omega}}{\omega_0^{d+1}} \left(\frac{\omega_0}{(\omega - \bar{x})^2} \right)^{\Delta_1} \left(\frac{\omega_0}{(\omega - \bar{y})^2} \right)^{\Delta_2} \left(\frac{\omega_0}{(\omega - \bar{z})^2} \right)^{\Delta_3} \Big|_{z \rightarrow 0}$$

$$= \frac{1}{|z_1|^{2\Delta} |z_2|^{2\Delta}} \int \frac{d\omega_0' d^d \bar{\omega}'}{(\omega_0')^{d+1}} \left(\frac{\omega_0'}{(\omega' - \bar{x}')^2} \right)^{\Delta_1} \left(\frac{\omega_0'}{(\omega' - \bar{y}')^2} \right)^{\Delta_2} (\omega_0')^{\Delta_3}$$

$$= \frac{1}{|z_1|^{2\Delta} |z_2|^{2\Delta}} \int \frac{d^{d+1}\omega'}{\omega'^2} \frac{(\omega_0')^{\Delta_1 + \Delta_2 + \Delta_3 - d - 1}}{|(\omega' - \bar{x}')^{2\Delta_1} (\omega' - \bar{y}')^{2\Delta_2}|}$$

$$\Gamma(\alpha) = \int_0^\infty dt t^{\alpha-1} e^{-t}$$

$$\Delta_1 + \Delta_2 + \Delta_3 = d + 1$$

$$\Gamma(\alpha) = \int_0^\infty A dz A^{\alpha-1} z^{\alpha-1} e^{-Az}$$

$$\Rightarrow \frac{\Gamma(\alpha)}{A^\alpha} = \int_0^\infty dt t^{\alpha-1} e^{-At}$$

Note:

$$\frac{1}{A_1^{\alpha_1} \dots A_n^{\alpha_n}} = \frac{\Gamma(\sum_i \alpha_i)}{\prod_i \Gamma(\alpha_i)} \frac{1}{(n-1)!} \int dF_n \frac{\prod_i x_i^{\alpha_i-1}}{(\sum_i x_i A_i)^{\sum_i \alpha_i}}$$

$$\frac{\Gamma(2(\Delta_1 + \Delta_2))}{\Gamma(2\Delta_1) \Gamma(2\Delta_2)} \frac{1}{(n-1)!} \int_0^1 d\alpha \frac{\alpha^{2\Delta_1-1} (1-\alpha)^{2\Delta_2-1}}{(\alpha(\omega_0^2 + (\omega-x)^2) + (1-\alpha)(\omega_0^2 + (\omega-y)^2))^{2(\Delta_1 + \Delta_2)}}$$

$$\bullet \alpha(\omega_0^2 + \bar{\omega}^2 - 2\bar{\omega}\bar{x} + \bar{x}^2)$$

$$= (\omega^2) - 2\alpha\bar{\omega}\bar{x} + \alpha\bar{x}^2 - (1-\alpha)2\bar{\omega}\bar{y} + (1-\alpha)\bar{y}^2$$

$$\bullet (\omega - \alpha\bar{x} - (1-\alpha)\bar{y})^2$$

$$= \omega_0^2 + (\bar{\omega} - \alpha\bar{x} - (1-\alpha)\bar{y})^2 \quad (\bar{\omega} - \alpha\bar{x} - (1-\alpha)\bar{y})$$

$$= \omega_0^2 + \bar{\omega}^2 - 2\alpha\bar{\omega}\bar{x} - 2(1-\alpha)\bar{\omega}\bar{y} + \alpha^2\bar{x}^2 + 2\alpha(1-\alpha)\bar{x}\bar{y} + (1-\alpha)^2\bar{y}^2$$

$$= (\omega - \alpha\bar{x} - (1-\alpha)\bar{y})^2 + \alpha\bar{x}^2 + (1-\alpha)\bar{y}^2 - 2\alpha(1-\alpha)\bar{x}\bar{y}$$

$$= \alpha^2\bar{x}^2 - (1-\alpha)^2\bar{y}^2$$

$$= \bar{x}^2(\alpha - \alpha^2) + \bar{y}^2(1-\alpha) - (1-\alpha)^2 - 2\alpha(1-\alpha)\bar{x}\bar{y}$$

$$= \bar{x}^2\alpha(1-\alpha) - \bar{y}^2(1-\alpha)(\alpha) + \alpha(1-\alpha)\bar{x}\bar{y}$$

$$= \alpha(1-\alpha)(\bar{x}^2 + \bar{y}^2 - 2\bar{x}\bar{y}) = \alpha(1-\alpha)(\bar{x} - \bar{y})^2$$

$$= (\underbrace{\omega - \alpha\bar{x} - (1-\alpha)\bar{y}}_{=q})^2 + \alpha(1-\alpha)(\bar{x} - \bar{y})^2 \stackrel{=}{=} D + q_0^2$$

$$\frac{\Gamma(2(\Delta_1 + \Delta_2))}{\Gamma(2\Delta_1) \Gamma(2\Delta_2)} \int_0^1 d\alpha \frac{\alpha^{2\Delta_1-1} (1-\alpha)^{2\Delta_2-1}}{(q^2 + D)^{2(\Delta_1 + \Delta_2)}}$$

$$\int \frac{d^d q_0}{(q_0^2 + D)^{d+1}} \left(\frac{\Gamma(1)}{\Gamma(D)} \right) \int_0^1 d\alpha \alpha^{2\Delta_1-1} \dots \int \frac{d^d \bar{q}}{(\bar{q}^2 + D)^{2(\Delta_1 + \Delta_2)}}$$

$$= \int \frac{\Gamma(2(\Delta_1 + \Delta_2) - \frac{1}{2}d)}{(4\pi)^{d/2} \Gamma(2(\Delta_1 + \Delta_2))} D^{-(2(\Delta_1 + \Delta_2) - \frac{d}{2})} \dots$$

$$= \underbrace{\int \frac{\Gamma(2(\Delta_1 + \Delta_2)) \Gamma(2(\Delta_1 + \Delta_2) - \frac{1}{2}d)}{\Gamma(2\Delta_1) \Gamma(2\Delta_2) (4\pi)^{d/2} \Gamma(2(\Delta_1 + \Delta_2))}}_{\beta} \left(\alpha(1-\alpha) |\bar{x} - \bar{y}|^2 + q_0^2 \right)$$

$$= (\beta - i) \int_0^1 d\alpha \alpha^{2\Delta_1-1} (1-\alpha)^{2\Delta_2-1} \int_{q_0^2}^{d\pi - 4\Delta_1\Delta_2} \frac{dq_0}{\left(\alpha(1-\alpha) |\bar{x} - \bar{y}|^2 + q_0^2 \right)^{-(2(\Delta_1 + \Delta_2) - \frac{d}{2})}}$$

$$= \beta(-i) \int d\alpha \dots \int \frac{dq_0}{\dots} \int_0^1 d\gamma \frac{\Gamma(d+1-\Delta_1-\Delta_2-\Delta_3+2\Delta_1+\Delta_2) - d/2}{\Gamma(d+1-\Delta_1-\Delta_2) \Gamma(2(\Delta_1+\Delta_2) - \frac{d}{2})}$$

$$= \widetilde{\beta}(-i) \left(d\alpha \dots \int_0^1 d\gamma \gamma^{\dots} (1-\gamma)^{\dots} \right)$$

$$\left(q_0^2 + (1-\gamma)(\alpha(1-\alpha) |\bar{x} - \bar{y}|^2 + q_0^2) \right)^{\dots}$$

$$\int_0^\infty dq_0 q_0^a (q_0^2 + D)^{-b} = \frac{1}{2} D^{\frac{a+1}{2}-b} \frac{\Gamma(\frac{a+1}{2}) \Gamma(b - \frac{a+1}{2})}{\Gamma^2(b)}$$

$$q_0^2 = Dt^2$$

$$dq_0 = t D dt$$

$$\rightarrow \int dt t D^{(a+1)/2} t^a D^{-b} (t^2 + 1)^{-b}$$

$$\int dt D^{(a+1)/2-b} t^{a-1} (t^2 + 1)$$

$$\int_0^\infty dt t^{-x-1} (1+t)^{-x-y} = \frac{\Gamma(x) \Gamma(y)}{\Gamma(x+y)}$$

$$u = t^2 \Rightarrow \frac{du}{2} = t dt$$

$$\frac{1}{2} \int du \left(u^{\frac{a-1}{2}-b} \right) u^{\frac{a-2}{2}} (u+1)^{-b}$$

$$u \rightarrow u-1$$

$$\frac{1}{2} \int du D (u-1)^{\frac{a-2}{2}+1-1} (u)^{-b}$$

$$= \frac{1}{2} D$$

$$= -i \int \frac{d^{d+1} \omega'}{(\omega_0')^2} \left(\frac{\omega_0}{\omega'} \right) \frac{1}{|x|^{2\Delta}} \frac{1}{|y|^{2\Delta}} (D) \left[\frac{(\bar{\omega}' - \bar{x}') i}{(\omega_0'^2 + (\bar{\omega}' - \bar{x}')^2)^2} K_{\Delta}(\omega', x') K_{\Delta}(\omega', y') \right. \\ \left. - \frac{(\bar{\omega}' - \bar{y}')}{(\omega_0'^2 + (\bar{\omega}' - \bar{y}')^2)^2} K_{\Delta}(\omega', x') K_{\Delta}(\omega', y') \right]$$

$$= (-i) D \int \frac{d^{d+1} \omega}{(\omega_0')^2} \frac{\omega_0^{2\Delta}}{(\omega')^2} \frac{1}{(\omega' - \bar{x})^{2\Delta}} \frac{1}{(\omega' - \bar{y})^{2\Delta}} \left[\right]$$

$$(A) \int \frac{d\omega_0' d\bar{\omega}'}{(\omega')^2} \omega_0^{2\Delta-1} \frac{2}{(\omega' - \bar{x})^{2\Delta}} \frac{1}{(\omega' - \bar{y})^{2\Delta}} \left(\frac{(\bar{\omega}' - \bar{x}') i}{\omega - \bar{x}} \right)$$

$$\int \frac{d\omega_0' d\bar{\omega}'}{(\omega')^2} \frac{\omega_0^{2\Delta-1}}{(\omega' - \bar{x})^{2\Delta+4}} \frac{1}{(\omega' - \bar{y})^{2\Delta}} (\bar{\omega}' - \bar{x}) i$$

$$= \frac{\Gamma(2+2\Delta+4+2\Delta)}{\Gamma(2)\Gamma(2\Delta+4)\Gamma(2\Delta)} \int_0^1 dx_1$$

* Derivative:

$$\frac{\partial}{\partial \omega_i'} \frac{\omega_0^{\Delta}}{(\omega_0'^2 + (\bar{\omega} - \bar{x})^2)^{\Delta}} = \frac{-\Delta 2 \omega_0^{\Delta} (\bar{\omega}_i - \bar{x}_i)}{(\omega_0'^2 + (\bar{\omega} - \bar{x})^2)^{\Delta+1}}$$

$$\rightarrow \langle J_i(\omega) \mathcal{O}_{\Delta}(u) \mathcal{O}_{\Delta}^*(v) \rangle$$

$$\int \frac{d\omega_0' d\bar{\omega}'}{(\omega')^{d+1}} (\omega_0')^{2\Delta} (\omega_0')^{d-1} \omega_0' (-2\Delta) \frac{(\bar{\omega}' - \bar{x}_i)}{(\omega_0'^2 + (\bar{\omega} - \bar{x})^2)^{\Delta+1}}$$

$\int \omega_0$

$$f_i 2\Delta \int \omega_0' d^d \bar{\omega}' \frac{\omega_0^{2\Delta-1} (\bar{X}_i - \gamma_i)}{(\omega_0^2 + (\bar{\omega} - \bar{x})^2)^{\Delta+1} (\omega_0^2 + (\bar{\omega} - \gamma)^2)^\Delta}$$

$$= -i 2\Delta$$

